Part I Report

Functional Module Discovery in Bacterial STRING Networks

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1. Introduction and Objective

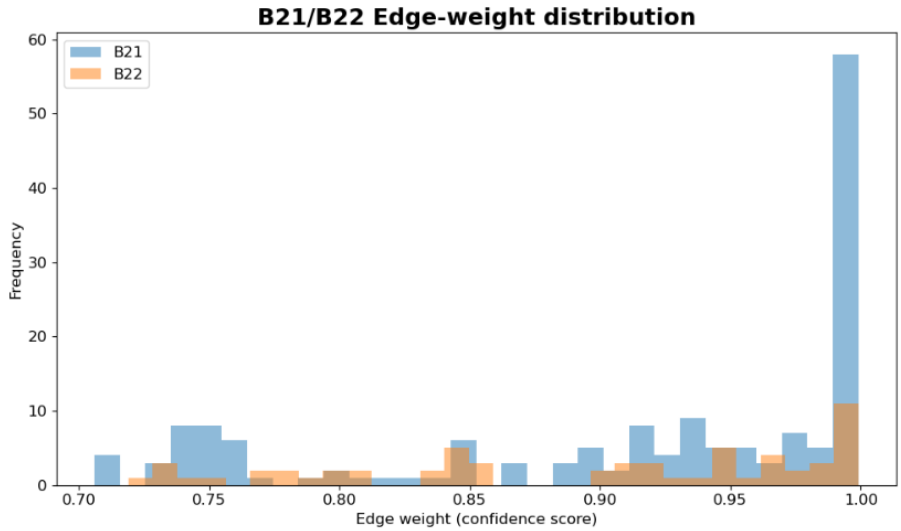
The aim of this part of the project was to discover modules within bacterial STRING-derived protein interaction networks and to assign them biological meaning using Gene Ontology (GO) enrichment. A module is defined as a densely connected group of proteins that likely corresponds to a shared biological process. We analyzed four bacterial datasets: two virulence networks (Salmonella virulence B5 and Vibrio cholerae virulence B6) and two resistance networks (Bacillus subtilis resistance B21 and Erwinia resistance B22). For each network we identified communities using a community-detection algorithm and then interpreted each module through enriched GO terms.

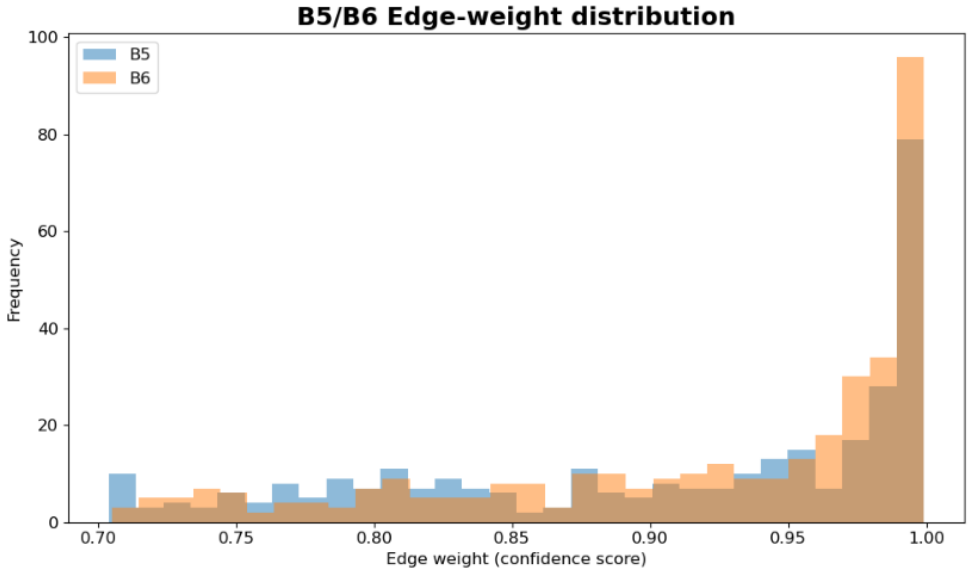
2. Data and Preprocessing

The input comprised four protein–protein interaction networks from STRING, each accompanied by functional annotation files. For every organism we constructed an undirected weighted graph where nodes represent proteins and edge weights correspond to the combined STRING confidence score. To reduce noise and ensure reliable modules, interactions with edge scores below 0.7 were discarded. Functional annotations mapped proteins to GO terms and served as the background for enrichment analysis.

Table 1. Summary of analyzed datasets.

	Input	Bacterium	Proteins	Interactions	Weighted	Annotation source
0	Input 1	B5	45	317	Yes	string_functional_annotations_5.tsv
1	Input 2	B6	68	356	Yes	string_functional_annotations_6.tsv
2	Input 3	B21	73	160	Yes	string_functional_annotations_21.tsv
3	Input 4	B22	30	61	Yes	string_functional_annotations_22.tsv





3. Module Detection Strategy

Several community detection methods were considered, including: Greedy Modularity, Spectral Clustering, Label Propagation, KMeans on adjacency.
Community detection approaches yielded higher modularity and more balanced partitions than general clustering algorithms. The Louvain algorithm was selected as the final method because it operates directly on weighted graph structure, supports undirected networks and produces compact, biologically meaningful modules. Parameter exploration indicated that an edge-score threshold of 0.7 yielded reliable modules across organisms. A minimum module size of 2 was used to preserve small yet functionally coherent communities. Stability tests with different random seeds showed high reproducibility, so a fixed seed was employed.

The final module detection parameters were:

Method: Louvain community detection
Edge-weight threshold: 0.7
Resolution/clustering parameter: default
Minimum module size: 2
Other relevant cutoff(s): random seed = 42

4. Functional Annotation and Enrichment Analysis

After detecting modules, we functionally characterized each one using Gene Ontology (GO) terms from all three categories: Biological Process, Molecular Function, and Cellular Component. For each module, we tested whether specific functional terms were over-represented relative to the background set of proteins in the corresponding bacterium.

We used the hypergeometric test for enrichment analysis and applied the Benjamini–Hochberg (FDR) correction method to control false positives. Terms were considered significant if they satisfied the following criteria:

Adjusted p-value or FDR cutoff: 0.05

Minimum overlap / minimum number of proteins per term: 2

Any additional filtering criterion: only modules with at least one enriched GO term were retained for further analysis.

5. Results

5.1 Overview of detected modules

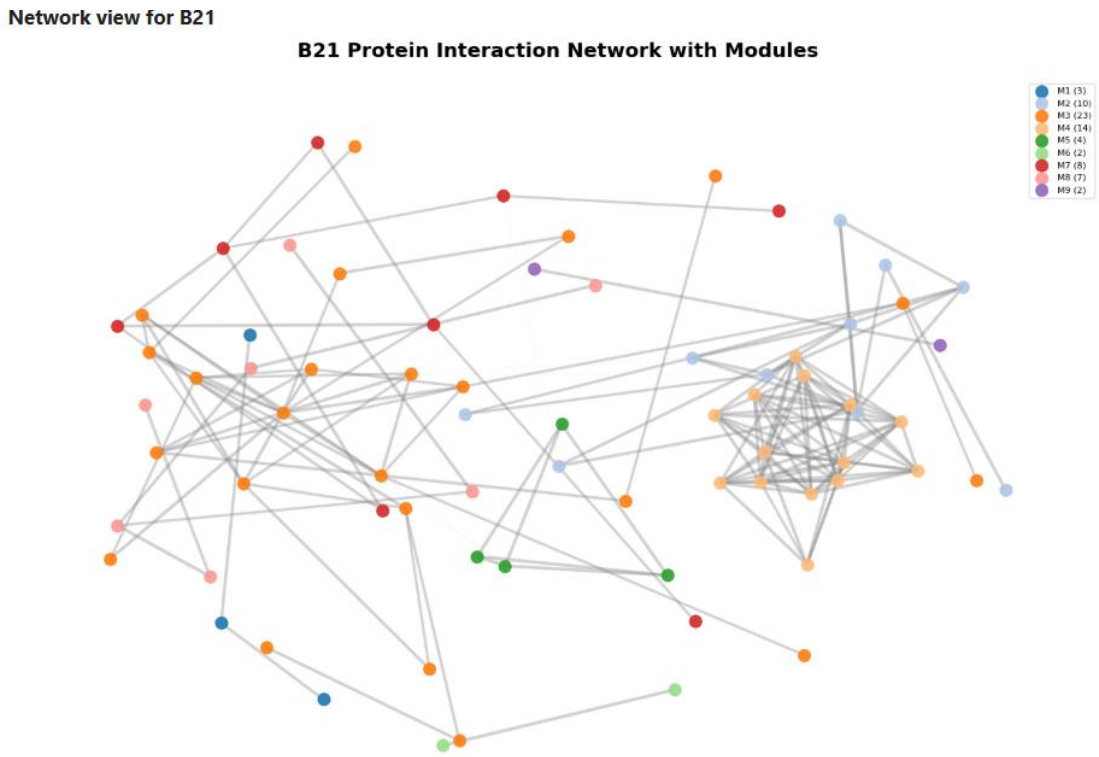
Table 2. Summary of detected modules across the four bacterial networks.

	Input	Bacterium	Number of modules	Largest module size	Smallest reported module size	Mean module size	Notes
0	Input 1	B5	4	17	6	11.25	
1	Input 2	B6	6	24	2	11.33	
2	Input 3	B21	9	23	2	8.11	
3	Input 4	B22	6	10	2	5.00	

Overall, the detected modules varied in size and density across the four networks. In general, modules corresponded to secretion and extracellular localization, biofilm formation and motility-related processes, toxin activity, type II secretion systems, transport and protein-localization modules, antibiotic biosynthetic pathways, carbohydrate/lipooligosaccharide biosynthesis, and transmembrane transporter systems.

5.2 Graphical representation of modules:

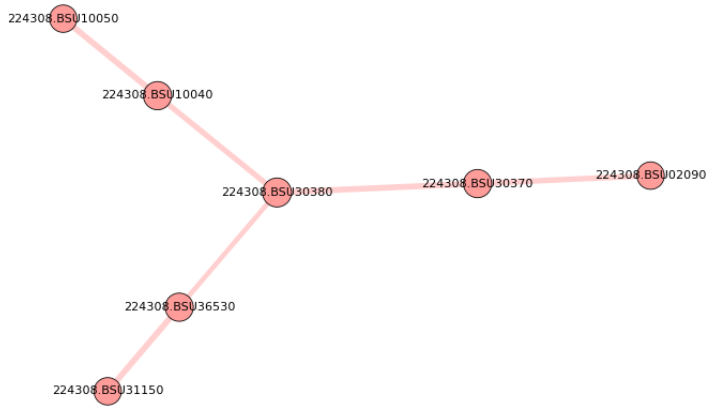
For each of the four inputs, we generated a graphical representation of the detected modules. In these figures, nodes represent proteins, edges represent STRING interactions, and colors indicate module membership.



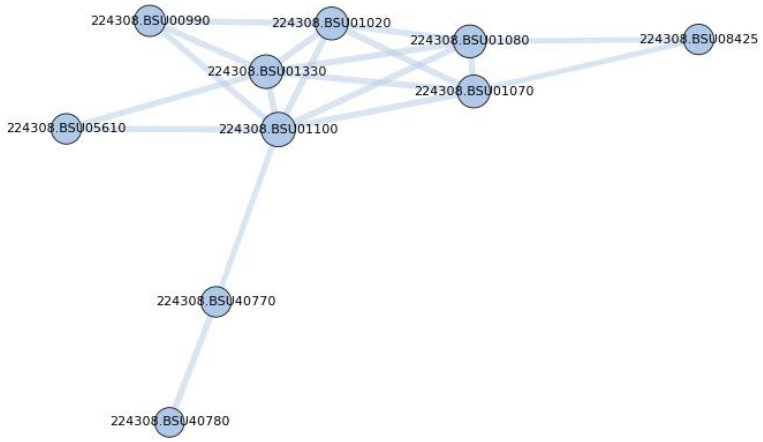
	bacterium	module_id	num_nodes	num_edges	density	num_significant_terms	best_term_id	best_term_description	best_term_category	best_p_value	best_fdr
0	B21	B21_M7	3	2	0.666667	56	GO:0070008	Serine-type exopeptidase activity	GO Function	1.607820e-05	3.657791e-04
1	B21	B21_M3	10	19	0.422222	52	GO:0010467	Gene expression	GO Process	2.443346e-06	3.274084e-04
2	B21	B21_M1	23	40	0.158103	45	GO:0050794	Regulation of cellular process	GO Process	3.799576e-07	1.048683e-04
3	B21	B21_M2	14	71	0.780220	34	GO:0045184	Establishment of protein localization	GO Process	1.622798e-13	1.805363e-12
4	B21	B21_M6	4	6	1.000000	14	GO:0017000	Antibiotic biosynthetic process	GO Process	9.187545e-07	4.777524e-05
5	B21	B21_M9	2	1	1.000000	5	GO:0140359	ABC-type transporter activity	GO Function	1.141553e-03	1.227169e-02
6	B21	B21_M4	8	9	0.321429	4	GO:0042221	Response to chemical	GO Process	2.312212e-04	8.092743e-03

B21_M8 | nodes=2 | edges=1

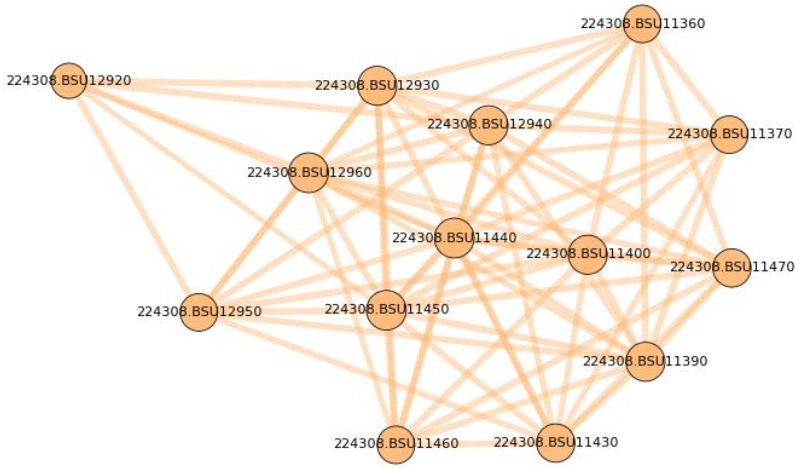
B21_M5 | nodes=7 | edges=6



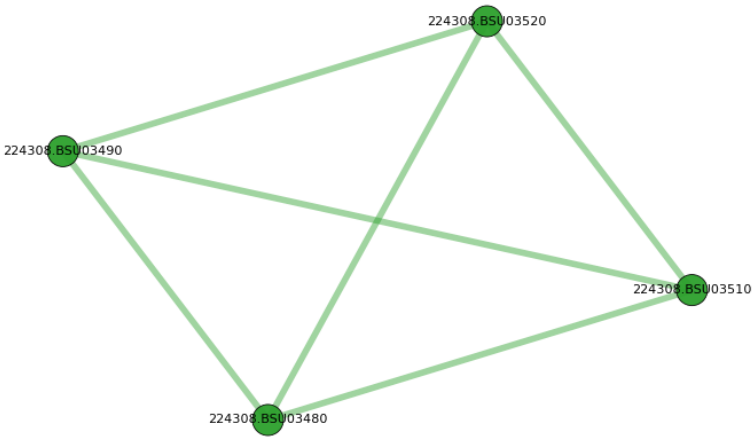
B21_M3 | nodes=10 | edges=19



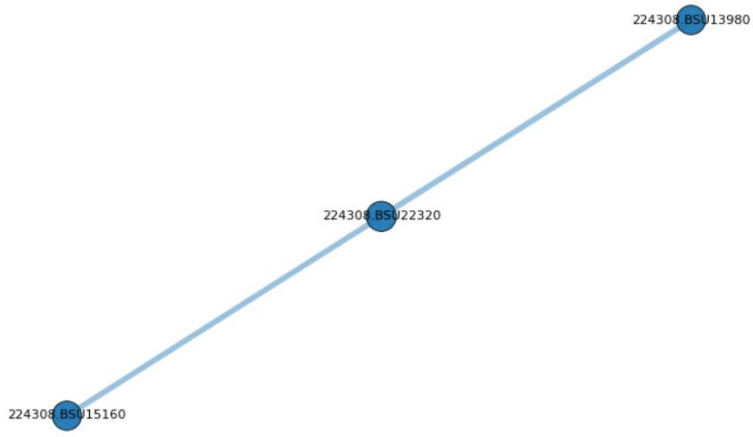
B21_M2 | nodes=14 | edges=71



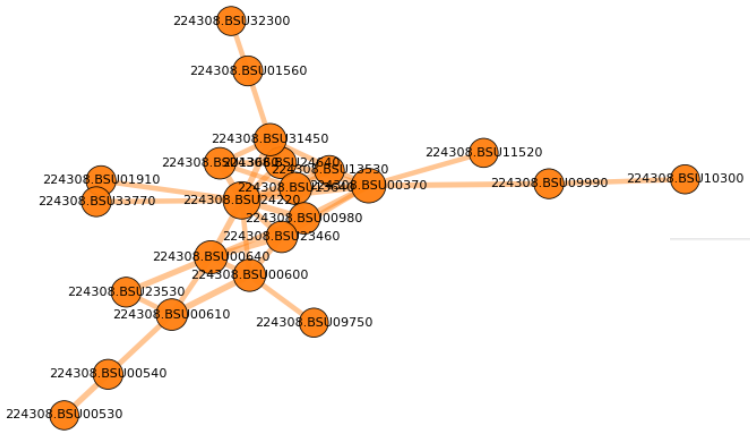
B21_M6 | nodes=4 | edges=6



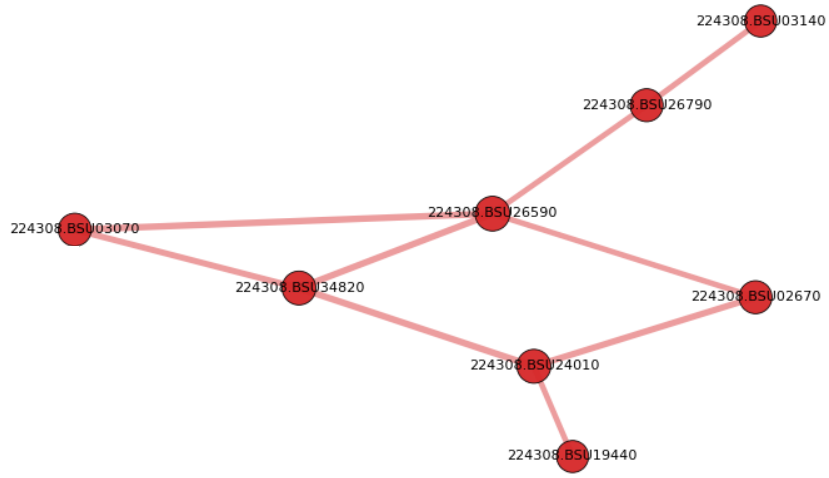
B21_M7 | nodes=3 | edges=2



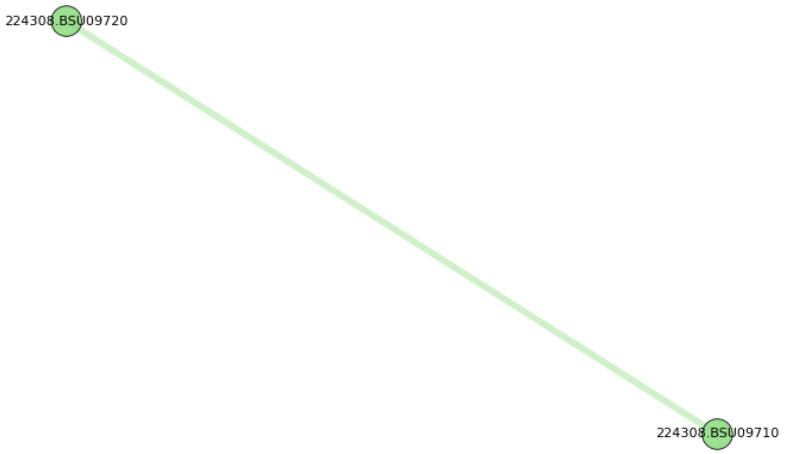
B21_M1 | nodes=23 | edges=40



B21_M4 | nodes=8 | edges=9

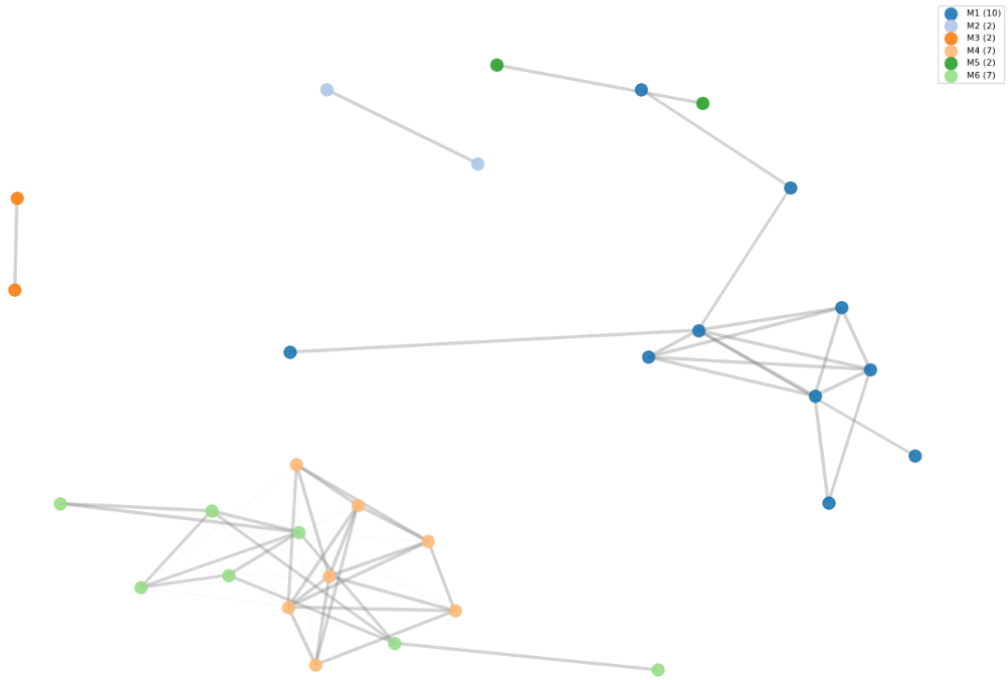


B21_M9 | nodes=2 | edges=1



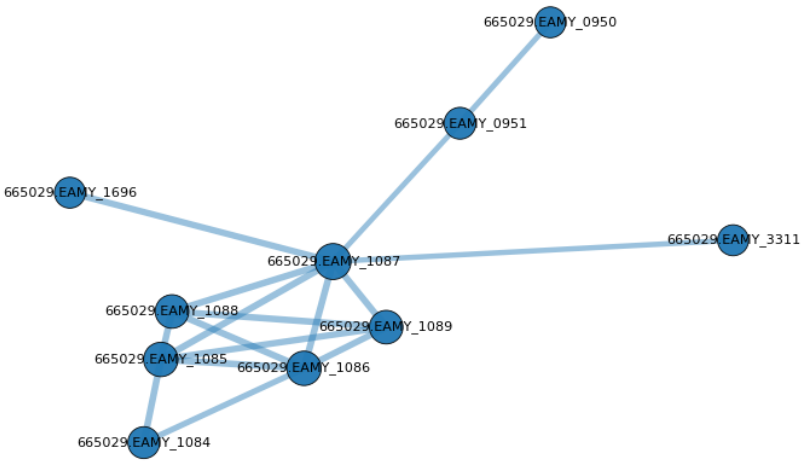
Network view for B22

B22 Protein Interaction Network with Modules

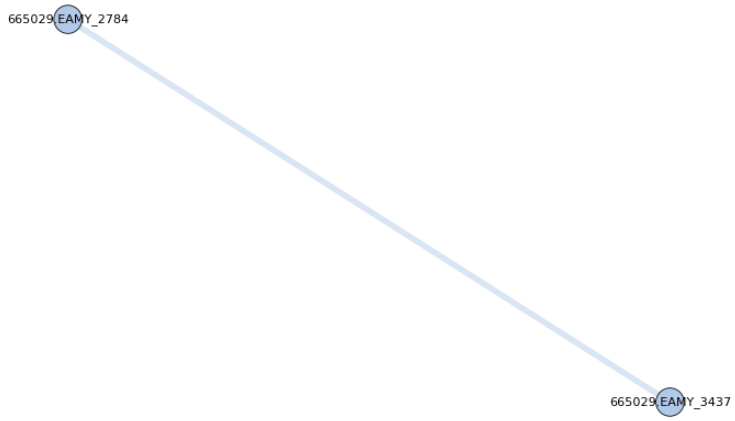


	bacterium	module_id	num_nodes	num_edges	density	num_significant_terms	best_term_id	best_term_description	best_term_category	best_p_value	best_fdr
0	B22	B22_M1	10	16	0.355556	56	GO:0009311	Oligosaccharide metabolic process	GO Process	9.985017e-08	3.860873e-07
1	B22	B22_M6	2	1	1.000000	44	GO:0008238	Exopeptidase activity	GO Function	2.463054e-03	5.867864e-03
2	B22	B22_M5	2	1	1.000000	18	GO:0015695	Organic cation transport	GO Process	2.463054e-03	5.418719e-03
3	B22	B22_M3	7	17	0.809524	6	GO:0055085	Transmembrane transport	GO Process	1.246044e-02	3.945805e-02
4	B22	B22_M4	2	1	1.000000	5	GO:0071236	Cellular response to antibiotic	GO Process	2.463054e-03	6.773399e-03

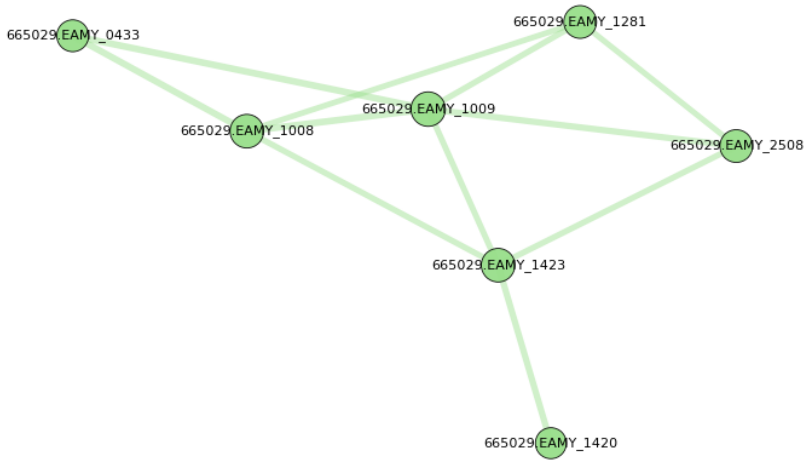
B22_M1 | nodes=10 | edges=16



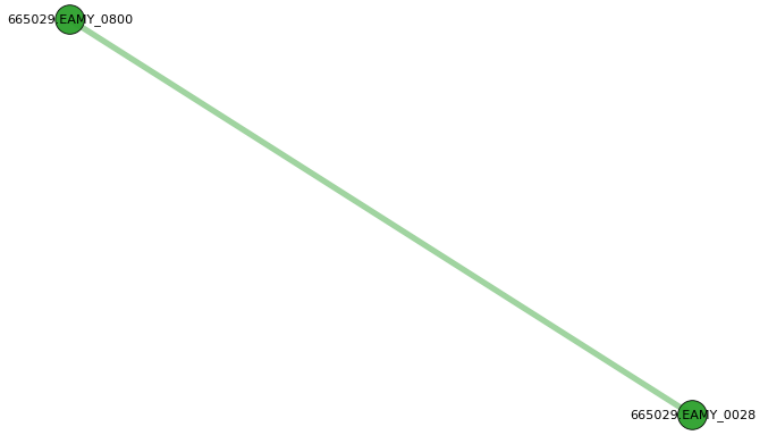
B22_M6 | nodes=2 | edges=1



B22_M2 | nodes=7 | edges=11

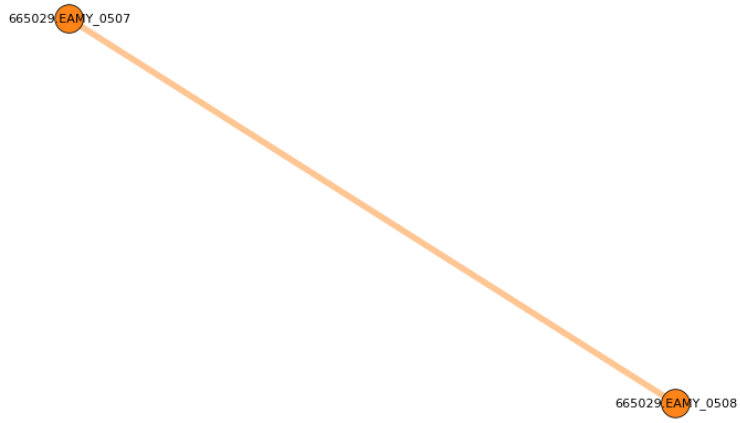


B22_M4 | nodes=2 | edges=1

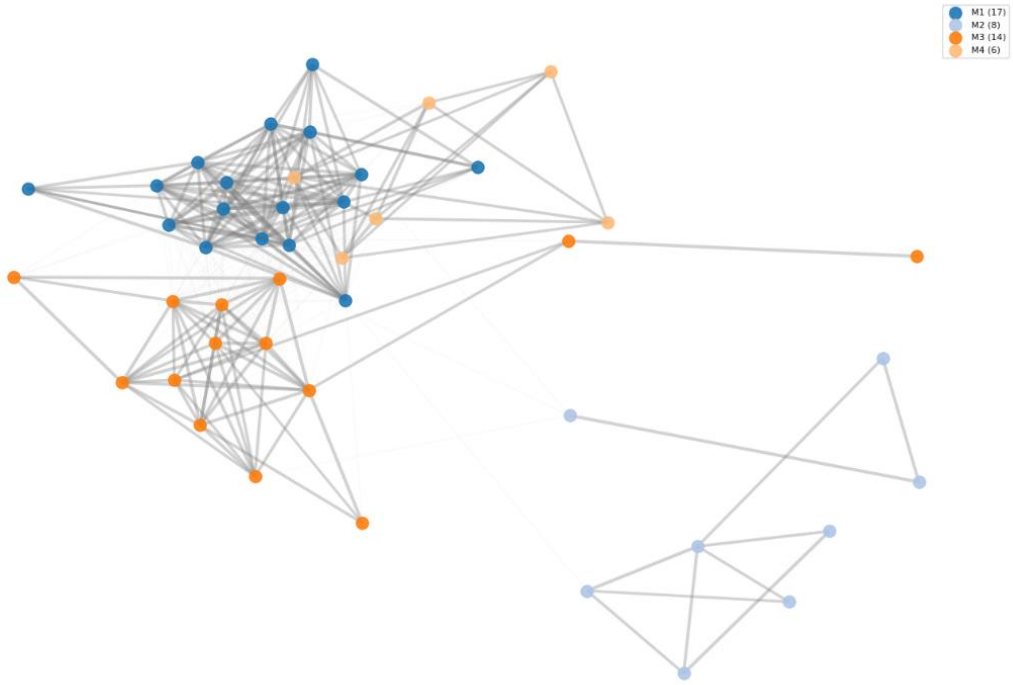


B22_M3 | nodes=7 | edges=17

B22_M5 | nodes=2 | edges=1

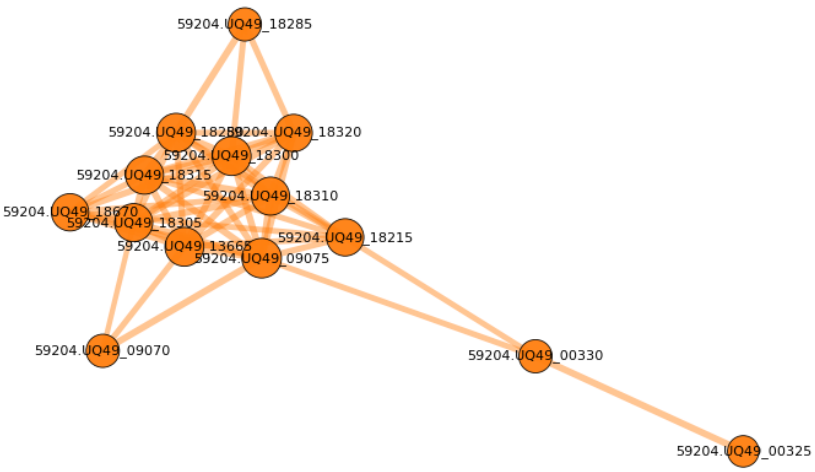


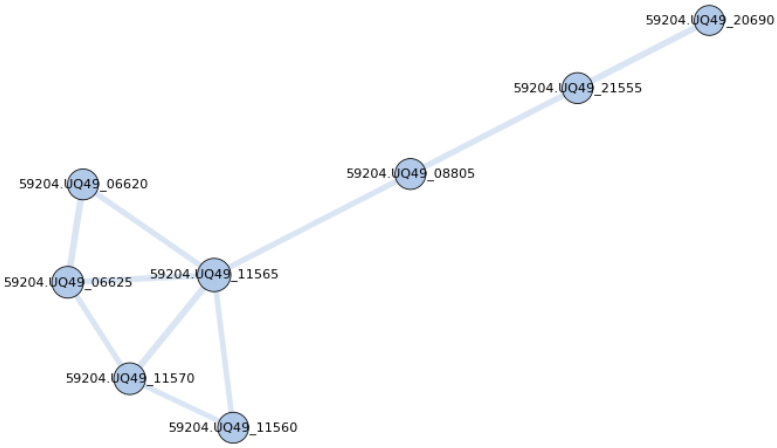
B5 Protein Interaction Network with Modules



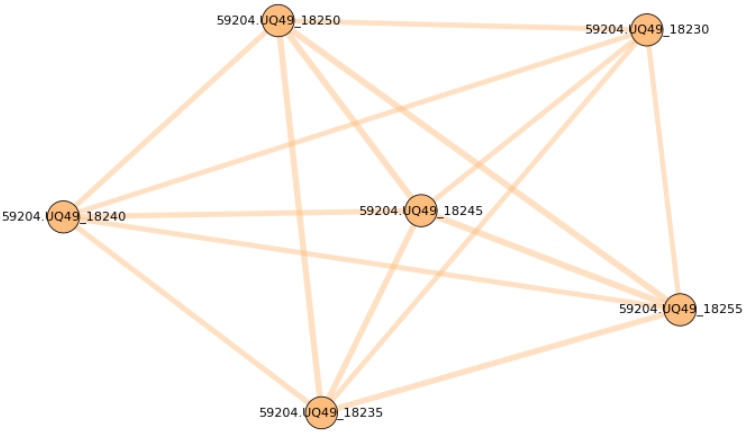
	bacterium	module_id	num_nodes	num_edges	density	num_significant_terms	best_term_id	best_term_description	best_term_category	best_p_value	best_fdr	
0	B5	B5_M1	17	110	0.808824	10	GO:0032940	Secretion by cell	GO Process	0.001379	0.036500	5
1	B5	B5_M3	8	10	0.357143	9	GO:0050896	Response to stimulus	GO Process	0.000075	0.007996	5
2	B5	B5_M2	14	51	0.560440	1	GO:0005576	Extracellular region	GO Component	0.000153	0.027783	5

B5_M2 | nodes=14 | edges=51

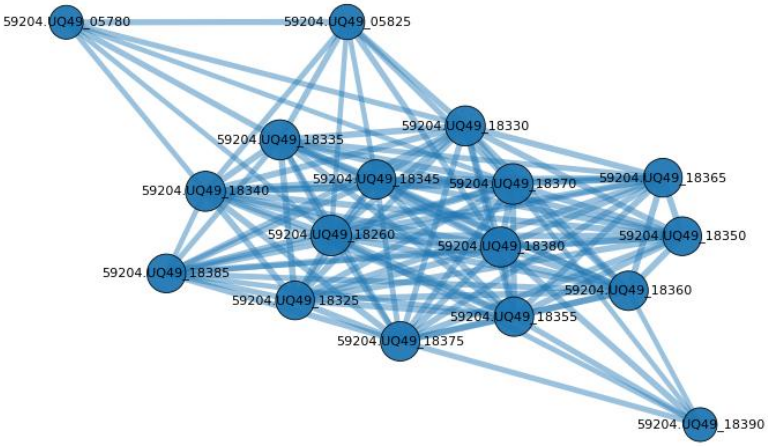




B5_M4 | nodes=6 | edges=15

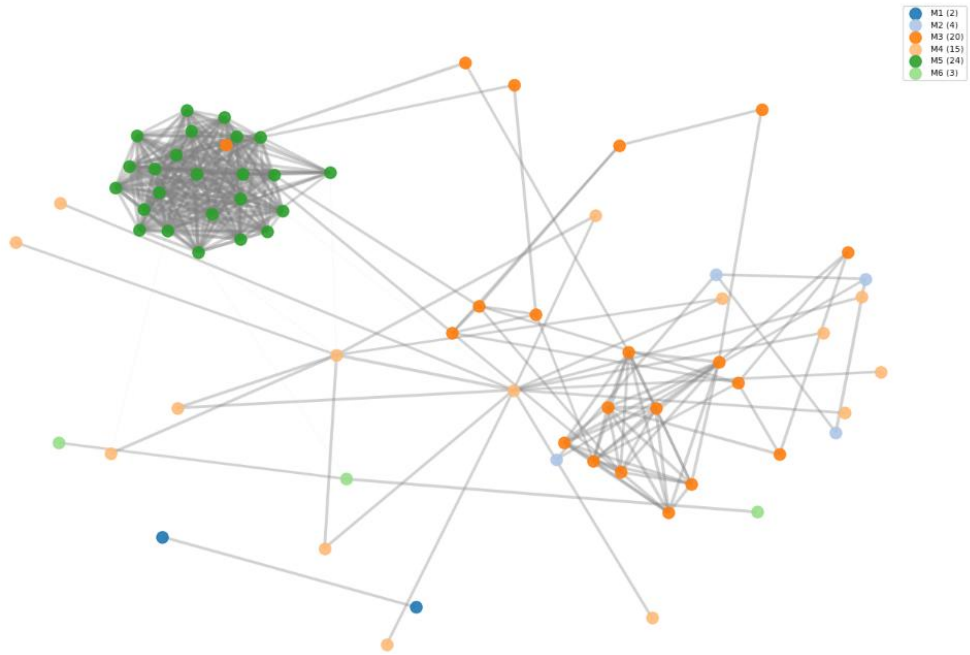


B5_M1 | nodes=17 | edges=110



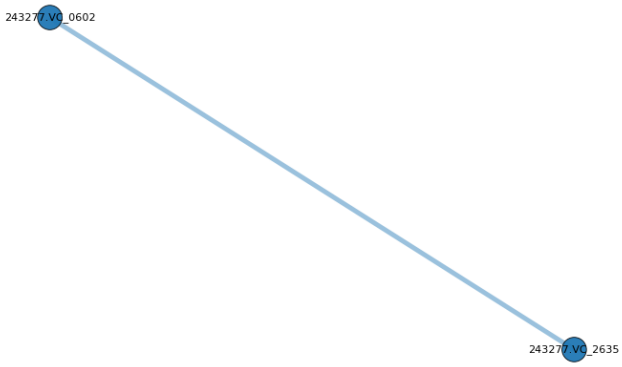
Network view for B6

B6 Protein Interaction Network with Modules

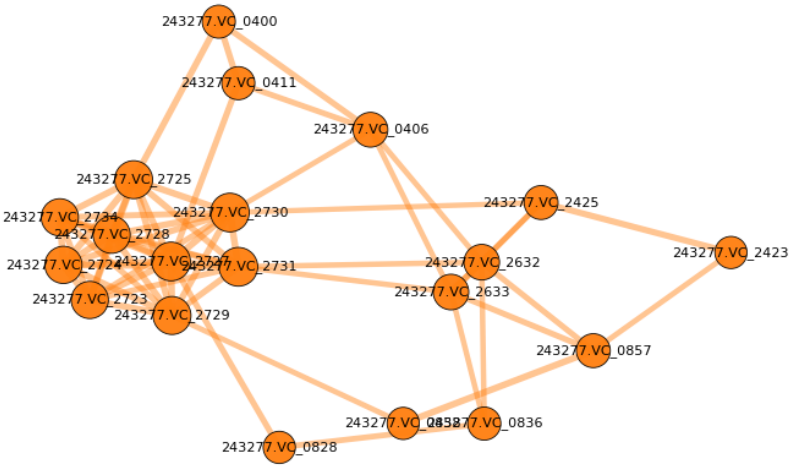


	bacterium	module_id	num_nodes	num_edges	density	num_significant_terms	best_term_id	best_term_description	best_term_category	best_p_value	best_fdr
0	B6	B6_M6	2	1	1.000000	67	GO:0008238	Exopeptidase activity	GO Function	4.389816e-04	2.370500e-03
1	B6	B6_M4	4	5	0.833333	42	GO:0044419	Biological process involved in interspecies in...	GO Process	1.841881e-05	3.438177e-04
2	B6	B6_M2	20	60	0.315789	31	GO:0015628	Protein secretion by the type II secretion system	GO Process	2.536066e-13	7.354591e-12
3	B6	B6_M3	15	18	0.171429	30	GO:0044010	Single-species biofilm formation	GO Process	2.254063e-15	5.747860e-15
4	B6	B6_M1	24	265	0.960145	16	GO:0097588	Archaeal or bacterial-type flagellum-dependent...	GO Process	6.650390e-19	6.949658e-17

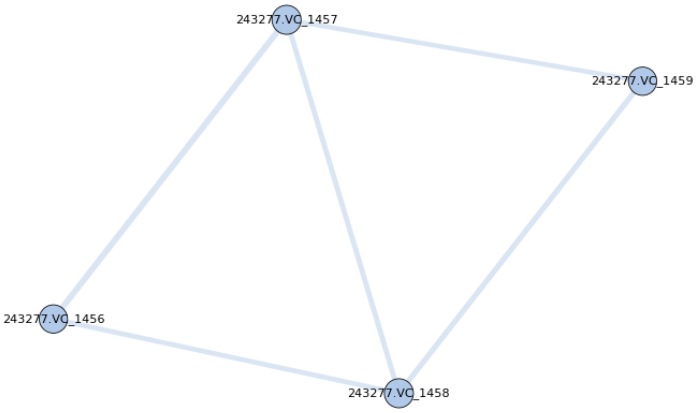
B6_M6 | nodes=2 | edges=1



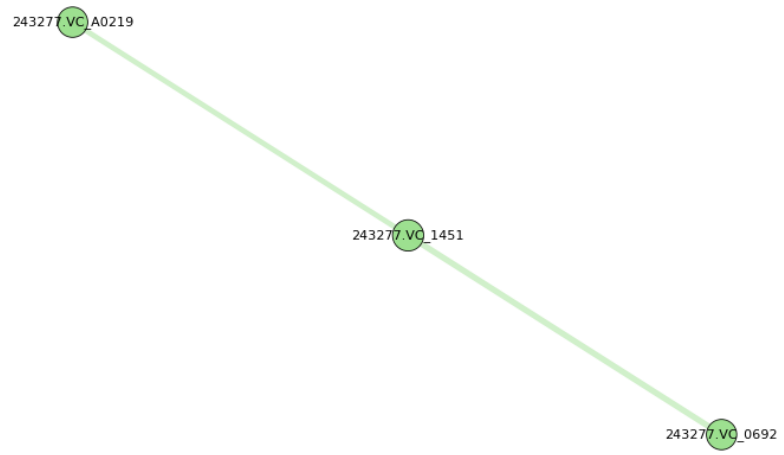
B6_M2 | nodes=20 | edges=60



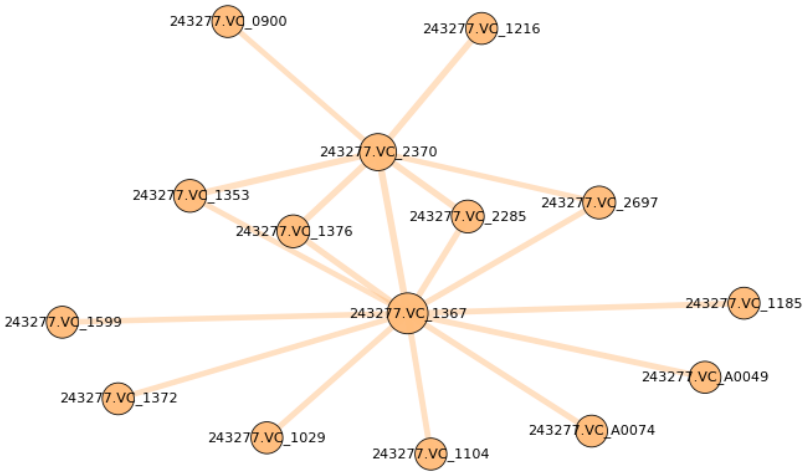
B6_M4 | nodes=4 | edges=5



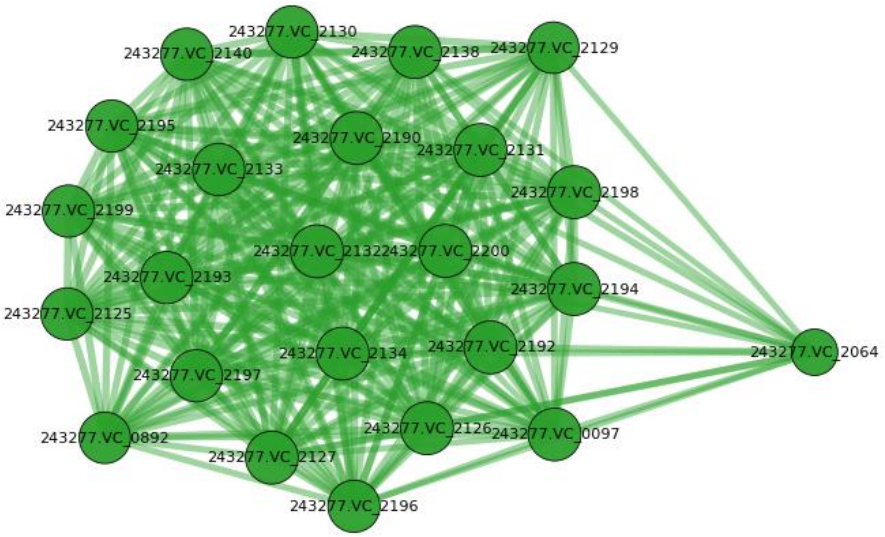
B6_M5 | nodes=3 | edges=2



B6_M3 | nodes=15 | edges=18



B6_M1 | nodes=24 | edges=265



5.3 Functional interpretation of modules

The GO enrichment analysis linked each detected module to coherent biological processes. Representative modules and their enriched terms are summarised in Table 3. These assignments highlight the diversity of biological functions captured by the detected communities.

Table 3. Functional summary of representative modules:

	Input	Bacterium	Module ID	Size	Top enriched term(s)	Functional interpretation
0	Input 1	B5	B5_M1	17	Secretion by cell, Protein secretion, Secretion	Secretion by cell
1	Input 1	B5	B5_M3	8	Response to stimulus, Regulation of RNA biosynthetic process, Regulation of biosynthetic process	Response to stimulus
2	Input 1	B5	B5_M2	14	Extracellular region	Extracellular region
3	Input 2	B6	B6_M6	2	Monocarboxylic acid binding, Serine-type carboxypeptidase activity, Sulfur compound binding	Monocarboxylic acid binding
4	Input 2	B6	B6_M4	4	Toxin activity, Modulation of process of another organism, Biological process involved in interspecies interaction between organisms	Toxin activity
5	Input 2	B6	B6_M2	20	Protein secretion by the type II secretion system, Protein transport across the cell outer membrane, Secretion	Protein secretion by the type II secretion system
6	Input 2	B6	B6_M3	15	Cell aggregation, Biofilm formation, Aggregation of unicellular organisms	Cell aggregation
7	Input 2	B6	B6_M1	24	Cilium or flagellum-dependent cell motility, Archaeal or bacterial-type flagellum-dependent cell motility, Bacterial-type flagellum-dependent cell motility	Cilium or flagellum-dependent cell motility
8	Input 1	B21	B21_M7	3	Serine-type exopeptidase activity, Carboxypeptidase activity, Serine-type D-Ala-D-Ala carboxypeptidase activity	Serine-type exopeptidase activity
9	Input 1	B21	B21_M3	10	Gene expression, Cellular nitrogen compound biosynthetic process, Response to chemical	Gene expression
10	Input 1	B21	B21_M1	23	Regulation of cellular process, Regulation of sporulation, Regulation of biological process	Regulation of cellular process
11	Input 1	B21	B21_M2	14	Protein transport, Nitrogen compound transport, Amide transport	Protein transport
12	Input 1	B21	B21_M6	4	Antibiotic biosynthetic process, Antibiotic metabolic process, Ligase activity	Antibiotic biosynthetic process
13	Input 1	B21	B21_M9	2	Primary active transmembrane transporter activity, ABC-type transporter activity, ATPase-coupled transmembrane transporter activity	Primary active transmembrane transporter activity
14	Input 1	B21	B21_M4	8	Response to chemical, Response to antibiotic, Transporter activity	Response to chemical
15	Input 2	B22	B22_M1	10	Membrane lipid metabolic process, Cellular polysaccharide metabolic process, Lipopolysaccharide metabolic process	Membrane lipid metabolic process
16	Input 2	B22	B22_M6	2	Monocarboxylic acid binding, Small molecule binding, Cellular component macromolecule biosynthetic process	Monocarboxylic acid binding
17	Input 2	B22	B22_M5	2	Ion transmembrane transporter activity, Spermidine transmembrane transporter activity, Ion transmembrane transport	Ion transmembrane transporter activity
18	Input 2	B22	B22_M3	7	Transmembrane transport, Establishment of localization, Transporter activity	Transmembrane transport
19	Input 2	B22	B22_M4	2	Cellular response to chemical stimulus, Periplasmic space, Cellular response to antibiotic	Cellular response to chemical stimulus

The enrichment analysis showed that many modules were associated with coherent biological functions. For example, modules enriched for **protein secretion by the type II secretion system** were interpreted as **secretion/export modules**, whereas modules enriched for **cilium or flagellum-dependent cell motility** or **submerged biofilm formation** were interpreted as **motility and biofilm-associated colonization modules**. Additional modules enriched for terms such as **transmembrane transporter activity**, **protein localization**, **antibiotic biosynthetic process**, and **oligosaccharide**

metabolic process were interpreted as **transport, localization, secondary-metabolism, and cell-envelope/carbohydrate biosynthesis functions**, respectively.

This supports the assumption that densely connected STRING subnetworks often correspond to biologically meaningful units. In line with these observations, a clear distinction was observed between the two types of input datasets. Modules identified in the resistance networks (B21, B22) were enriched for antibiotic-related processes, transport activity, and response to chemical stimuli, whereas modules in the virulence networks (B5, B6) were associated with secretion systems, toxin activity, and motility-related functions.

6. Discussion.

The selected module detection method was appropriate for this task because Louvain community detection operates directly on the weighted STRING graph, identifies groups with strong internal connectivity, does not require pre-specifying the number of modules, and produced biologically interpretable communities across the bacterial networks**.** Compared with the alternative approaches we considered, it provided a better balance between graph-structural relevance, interpretability, and robustness [As shown in the experiments part 1.14 Part 1 experiments].
Unlike k-means and spectral clustering, it did not require forcing the network into a fixed number of clusters, and unlike label-propagation-based approaches, it was more stable and easier to interpret biologically**.**

The main parameters that influenced the results were the edge-weight filtering threshold and the module-detection settings. In our pipeline, the interaction graph was filtered at a combined_score threshold of 0.7, so a stricter threshold would have produced a sparser graph with fewer but tighter connections, often splitting modules into smaller and more specific communities, whereas a more permissive threshold would have produced a denser graph with larger or partially merged modules, at the cost of including weaker and potentially noisier associations. Similarly, changing the clustering settings would affect the number and granularity of the detected modules.

From the enrichment side, the choice of significance cutoff and descriptor set also influenced interpretation. We used Benjamini-Hochberg FDR correction with $\alpha = 0.05$ and restricted the analysis to GO Process, GO Function, and GO Component terms. More stringent thresholds reduced noise but sometimes removed broader biological signals, while more permissive thresholds increased the number of reported terms but also the chance of weak or less informative enrichments. Overall, the final settings produced modules that were both topologically coherent and functionally interpretable.

In addition, the identified modules are consistent with the biological nature of the input datasets. In the resistance networks (B21, B22), several modules were enriched for processes such as antibiotic biosynthesis, transport activity, and response to chemical stimuli, reflecting known resistance mechanisms. In contrast, the virulence networks (B5, B6) showed enrichment for secretion systems, toxin-related activity, and motility-associated processes, which are characteristic of virulence and host interaction. This

agreement between network-derived modules and biological expectations further supports the validity of the analysis.

7. Output Files

The code is designed to read all input files from the current directory and save output files to the current directory.

Table 4. Output files generated for Part I:

	File name	Description
0	B21_B22_all_bacteria_module_summary.csv	Summary table of detected modules
1	B21_M1_enrichment.csv	CSV output file
2	B21_M2_enrichment.csv	CSV output file
3	B21_M3_enrichment.csv	CSV output file
4	B21_M4_enrichment.csv	CSV output file
5	B21_M5_enrichment.csv	CSV output file
6	B21_M6_enrichment.csv	CSV output file
7	B21_M7_enrichment.csv	CSV output file
8	B21_M8_enrichment.csv	CSV output file
9	B21_M9_enrichment.csv	CSV output file
10	B21_module_summary.csv	Summary table of detected modules
11	B21_module_terms_detailed.csv	CSV output file
12	B22_M1_enrichment.csv	CSV output file
13	B22_M2_enrichment.csv	CSV output file
14	B22_M3_enrichment.csv	CSV output file
15	B22_M4_enrichment.csv	CSV output file
16	B22_M5_enrichment.csv	CSV output file
17	B22_M6_enrichment.csv	CSV output file
18	B22_module_summary.csv	Summary table of detected modules
19	B22_module_terms_detailed.csv	CSV output file
20	B5_B6_all_bacteria_module_summary.csv	Summary table of detected modules
21	B5_M1_enrichment.csv	CSV output file
22	B5_M2_enrichment.csv	CSV output file
23	B5_M3_enrichment.csv	CSV output file
24	B5_M4_enrichment.csv	CSV output file
25	B5_module_summary.csv	Summary table of detected modules
26	B5_module_terms_detailed.csv	CSV output file
27	B6_M1_enrichment.csv	CSV output file
28	B6_M2_enrichment.csv	CSV output file
29	B6_M3_enrichment.csv	CSV output file
30	B6_M4_enrichment.csv	CSV output file
31	B6_M5_enrichment.csv	CSV output file
32	B6_M6_enrichment.csv	CSV output file
33	B6_module_summary.csv	Summary table of detected modules
34	B6_module_terms_detailed.csv	CSV output file

8. Conclusion

Part I established a reproducible pipeline for extracting biologically meaningful structure from the four bacterial STRING networks. By combining weighted graph preprocessing, community detection on the interaction topology, and GO-based enrichment analysis, we converted protein interaction patterns into biologically interpretable functional modules. The modules identified in this stage captured biological themes such as secretion-related systems, transport activity, motility and biofilm-associated behavior, toxin-linked functions, biosynthetic pathways, and cell-envelope or carbohydrate-associated processes

Across the four bacterial networks, the detected modules, derived from two virulence datasets (B5, B6) and two resistance datasets (B21, B22), were not random graph partitions but biologically coherent units.

This supports the idea that densely connected protein subnetworks in STRING often reflect shared biological roles. The resulting module catalogs, enrichment tables, and network visualizations will provide a foundation for the comparative module analysis performed in Part II, where functional conservation and divergence can be examined across and within species.